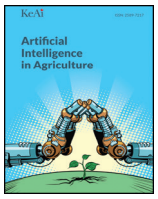
Artificial Intelligence in Agriculture xxx (xxxx) xxx

Contents lists available at ScienceDirect

Artificial Intelligence in Agriculture

journal homepage: http://www.keaipublishing.com/en/journals/artificial-intelligence-in-agriculture/



Research Paper

Integrating machine learning and large language models to enhance risk assessment and priority management for plant biosecurity

Haoxiang Zhao [a], Shanqing Yi [a], Andy Sheppard [b], Rieks van Klinken [b], Nianwan Yang [a,c], Jue Wang [d], Xiaoqing Xian [a,*], Fanghao Wan [a], Wanxue Liu [a,*]

[a] State Key Laboratory for Biology of Plant Diseases and Insect Pests, Key Laboratory for Prevention and Control of Invasive Alien Species of Ministry of Agriculture and Rural Affairs, Institute of Plant Protection, Chinese Academy of Agricultural Sciences, Beijing, China
[b] CSIRO Health & Biosecurity, Canberra, Australian Capital Territory, Australia
[c] Western Agricultural Research Center, Chinese Academy of Agricultural Sciences, Changji, China
[d] Smith School of Business, Queen's University, Kingston, Canada

ARTICLE INFO

Article history:
Received 4 March 2026
Received in revised form 23 July 2026
Accepted 29 July 2026
Available online xxxx

Keywords:
Citrus pest
Artificial intelligence
Multistage risk modeling
Decision-support system
Risk management framework

ABSTRACT

Global food security and biosecurity are continually threatened by the prevalence and spread of plant pests with the ever-accelerating rate of global trade. The continuous advancement of artificial intelligence, such as large language models (LLMs) and machine learning, provides novel insights into plant pest risk management in the food trade, whereas their practical effectiveness remains insufficiently understood. Here, by integrating an LLM with the retrieval-augmented generation method, a machine learning-based network analysis and ecological niche modeling, we constructed a risk management framework and a decision support system to assess the multistage risk of pests in the citrus trade, thereby supporting priority risk management. The introduction risk of citrus pests tended to increase from 2003 to 2018, with 243 introduction risk pathways and 98 risk nodes, mainly including scale insects and fruit flies. Our findings highlight the high risk of co-introduction and co-establishment of multiple citrus pests at important citrus planting areas in southern China, as well as the risk of spread into high-latitude regions under climate change. The decision support platform demonstrated reliable citrus pest risk management performance. Additionally, our study provides novel insights into multistage risk management of plant pests with various trade pathways.

© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

The prevalence of plant pests causes substantial economic losses in crop production scenarios and poses a persistent threat to food security (Maxmen, 2013; Wang et al., 2022). Approximately 40% of crop production worldwide is lost to plant pests (Gullino et al., 2021; Savary et al., 2019). With the acceleration of globalization and global trade, the transboundary spread of plant pests has significantly increased, leading to emerging food security and biosecurity risks, including biological invasions and the emergence of pest populations with novel resistance traits (Montgomery et al., 2023; Nardi et al., 2025; Tarusikirwa et al., 2022; Turner et al., 2021). Climate warming has further contributed to the expansion of pest distributions and the acceleration of their developmental cycles, resulting in more severe crop damage (Deutsch et al., 2018; Ma et al., 2025; Xiao et al., 2017). In response to these growing challenges, the Food and Agriculture Organization of the United Nations (FAO) developed the International Plant Protection Convention (IPPC) (IPPC, 2025), and the World Trade Organization (WTO) implemented the Agreement on the Application of Sanitary and Phytosanitary Measures (World Trade Organization, 1994) to manage the plant pest risks associated with global trade. Many countries and regions have developed quarantine pest lists to regulate the entry of restricted plant pests and mitigate the biosecurity risks that are associated with global

Abbreviations: FAO, Food and Agriculture Organization of the United Nations; IPPC, International Plant Protection Convention; WTO, World Trade Organization; LLM, Large Language Model; RAG, Retrieval-Augmented Generation; ENM, Ecological Niche Modeling; GBIF, Global Biodiversity Information Facility; CNKI, China National Knowledge Infrastructure; SSP, Shared Socioeconomic Pathway; GCM, General Circulation Model; HFP, Human Footprint; NPP, Net Primary Production; GLM, Generalized Linear Model; RF, Random Forest; GBM, Gradient-Boosting Machine; MaxEnt, Maximum Entropy Model; MARS, Multivariate Adaptive Regression Splines; AUC, Area Under the Receiver Operating Characteristic Curve; TSS, True Skill Statistic; ZAF, South Africa; HK, Hong Kong, China; MO, Macao, China; USA, United States; EGY, Egypt; VNM, Vietnam; TW, Taiwan, China; THA, Thailand; SGP, Singapore; BJ, Beijing; FJ, Fujian; GD, Guangdong; GS, Gansu; GX, Guangxi; GZ, Guizhou; HB, Hubei; HI, Hainan; HL, Heilongjiang; HN, Hunan; JS, Jiangsu; NX, Ningxia; QH, Qinghai; SC, Sichuan; SH, Shanghai; SN, Shaanxi; XJ, Xinjiang; YN, Yunnan; SZ, Shenzhen.

* Corresponding authors.
E-mail addresses: xianxiaoqing@caas.cn (X. Xian), liuwanxue@caas.cn (W. Liu).

https://doi.org/10.1016/j.aiia.2026.07.004
2589-7217/© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Please cite this article as: H. Zhao, S. Yi, A. Sheppard, et al., Integrating machine learning and large language models to enhance risk assessment and priority manag..., , https://doi.org/10.1016/j.aiia.2026.07.004

trade. For instance, the European and Mediterranean Plant Protection Organization (EPPO) formulated a quarantine pest list for the European Union (Mathys and Smith, 1984), and China issued a catalog of quarantine pests for imported plants (Sun et al., 2017) to safeguard national biosecurity and crop production.

The development of quantitative risk analysis methods for plant pests is essential for determining the associated multistage "unrestricted" risks in the food trade, that is, the pest introduction risks that are mediated via the commodity trade, as well as their establishment and spread risk into the relevant domestic production regions (Froese et al., 2024; Rosace et al., 2025; van Klinken et al., 2023), thereby supporting stage-specific risk management and surveillance planning processes. Such approaches can inform the creation and revision of quarantine pest lists for imported plants and support the development of targeted risk management strategies to reduce the threats caused by plant pests to food security and biosecurity. Numerous quantitative models for assessing pest risks in the food trade have been developed, most of which were designed for specific plant pest groups or particular risk stages. For instance, the integrated biosecurity risk assessment model can evaluate the multistage risks associated with the introduction, establishment, and spread of pests (Jamieson et al., 2022). However, its ability to provide practical support for the development of pest-specific, stage-targeted management strategies remains limited. Recent studies have integrated network analyses and ecological niche modeling to establish a coupled multistage risk analysis framework for mealybugs, addressing their introduction, establishment, and spread processes (Zhao et al., 2024a). Nevertheless, comprehensive frameworks that can integrate multiple pest taxa; multiple trade pathways; and the sequential stages of introduction, establishment, and spread into a unified management-oriented workflow are lacking. In particular, the existing approaches have only a limited ability to transform multistage quantitative risk outputs into transparent, evidence-based, and user-friendly decision support for regulators and quarantine managers. Strengthening this connection between quantitative risk assessment and management decision support is therefore essential for prioritizing high-risk pathways and taxa, guiding surveillance processes in vulnerable production regions, and improving stage-targeted phytosanitary interventions under intensifying global trade and climate change conditions.

Large language models (LLMs), which are based on the transformer architecture and its self-attention mechanism (Vaswani et al., 2017), have demonstrated remarkable knowledge emergence capabilities in terms of processing and reasoning over long textual information (Yin et al., 2024). These advances provide a novel perspective for addressing the complex challenges associated with plant pest risk management in the global food trade. LLMs have been applied in various scientific domains, including agricultural production through intelligent decision support and question-answering systems and pest control via multimodal diagnosis, pest identification, and interactive advisory applications (Tzachor et al., 2023; Wang et al., 2024; Yan et al., 2025). However, their utility and potential for managing the multistage risks brought by the introduction, establishment, and spread of plant pests within the diverse global food trade remain poorly explored. General-purpose LLMs possess robust general reasoning and linguistic capabilities but exhibit limited effectiveness in terms of supporting plant pest risk management in the food trade. This limitation primarily arises from their lack of domain-specific knowledge related to pest risks in the food trade (Li et al., 2025). Moreover, plant biosecurity risk management requires stage-specific introduction, establishment, and spread information. However, such information is often unavailable from the literature or online data sources, which frequently lack structured and taxon-specific evidence for particular plant pest groups across different risk stages. Consequently, the key risk information often needs to be generated through quantitative modeling. How to integrate these model-based risk predictions with LLMs to provide accurate and evidence-grounded decision support for managers and policymakers remains insufficiently understood. By integrating machine learning-based network analyses and ecological niche modeling (ENM), it is possible to perform quantitative multistage risk assessments of plant pests and generate high-quality datasets for risk management purposes (Nardi et al., 2025; Thuiller, 2024; Zhao et al., 2024a). Furthermore, incorporating the retrieval-augmented generation (RAG) framework can substantially increase the decision-support capacities of LLMs (Lewis et al., 2021; Li et al., 2025). This integrated approach provides novel insights into the development of advanced risk management approaches and decision support systems for managing the multistage risks posed by multiple plant pest taxa across various trade pathways.

Citrus species (Sapindales: Rutaceae) are the most widely cultivated fruits in the world and are planted in approximately 140 countries and regions; five dominant citrus fruits were considered in this study: oranges, lemons, mandarins, grapefruits, and pomelos (de Souza et al., 2020; Wu et al., 2018). Citrus species have also been widely planted in China, with a planting area of approximately 3.5 million hectares (Wei et al., 2025). Approximately 70 prevalent pests are present in the citrus planting areas in China, such as scale insects (Hemiptera: Coccoidea) and fruit flies (Diptera: Tephritidae), and they cause substantial economic losses to citrus production (Niu et al., 2014; Wei et al., 2025). Citrus species are also considered among the most important fruit categories in the food trade worldwide, as well as in the fruit import trade of China. Citrus species imports have exhibited a continual increasing trend in China since 2010, with an average annual increase of 23.42% from 2010 to 2019 (UN Comtrade, 2025). The ever-increasing citrus trade has facilitated the cross-border spread of citrus pests, posing threats and potential risks to citrus production in China. The interception frequency of citrus pests, such as scale insects and fruit flies, at China's entry ports has significantly increased, which is driven mainly by the growing trade in citrus, resulting in an increased risk of citrus pest introduction (Xu and Gu, 2023; Ye et al., 2017). However, the multistage risks associated with the introduction, establishment, and spread of citrus pests via various citrus species trade pathways remain unclear.

Here, using citrus pests as an example, our study integrated a machine learning-based network analysis and ecological niche modeling to estimate the multistage risks of citrus pests (Fig. 1); then, we generated decision support using the LLM and RAG methods to integrate all risk analysis knowledge-related data (Fig. 1). Specifically, our study had the following objectives: (i) identifying the high-risk introduction nodes (i.e., sources and entry ports), pathways, and citrus pests in the introduction risk networks mediated by trade in various citrus species to help prioritize pest prevention and quarantine strategies; (ii) estimating the establishment risks of all citrus pests (e.g., scale insects and fruit flies) in the citrus-growing regions of China and modeling their potential spread risk dynamics under future climate change conditions to develop effective surveillance and containment strategies; and (iii) constructing a multistage risk management framework and decision support system for citrus pests that integrates quantitative modeling and knowledge-based reasoning to enhance the scientific basis for updating quarantine pest lists, optimize biosecurity surveillance schemes, and support the development of risk management strategies in response to increasing global trade and climate change.

2. Material and methods

In this study, a multistage risk assessment framework was constructed to evaluate citrus pest risks along trade pathways, covering the introduction, establishment, and spread phases (Fig. 1). An introduction risk analysis was conducted using the interceptions of citrus pests to characterize pest introduction risks. Establishment and spread risks were subsequently assessed using species distributions and environmental data through ecological niche modeling to estimate potentially suitable areas for citrus pests. These complementary analyses were ultimately integrated into a unified risk management and decision

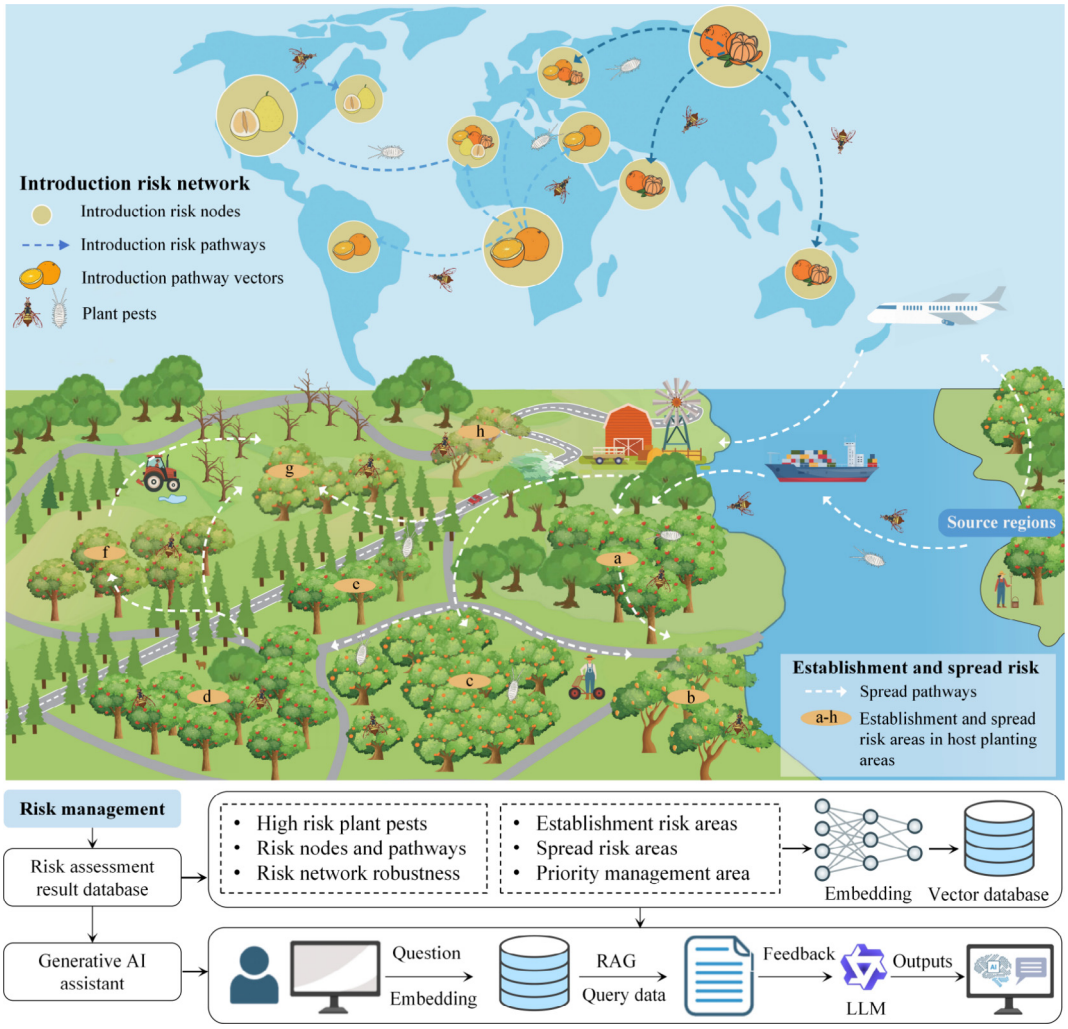


Fig. 1. Conceptual workflow integrating the generative AI and machine learning approaches for assessing and managing plant pest introduction, establishment, and spread risks. First, the introduction risk network is analyzed, including the introduction risk nodes, pathways, and commodities that yield plant pest vectors. Afterward, the establishment and spread risks of the plant pests located within host crop cultivation regions are quantified. Finally, by integrating the multistage risk assessment results (i.e., introduction, establishment and spread results), a vector database is constructed to support the development of a decision support system (based on LLMs and RAG), which can allow the system to quickly retrieve and relate relevant information across multiple risk stages, thereby facilitating the identification of high-risk plant pests, critical introduction pathways, and priority management areas. The map lines delineate the study areas and do not necessarily depict accepted national boundaries.

support platform based on a large language model with a retrieval-augmented generation (RAG) framework.

2.1. Interception, citrus import volume and yield data

Interception and citrus yield data were collected to characterize the citrus pest introduction risks by combining evidence of pest entry events with information concerning the scales of citrus production across different provinces. The interception data for citrus pests from 2003 to 2018 included their source countries or regions, entry ports within China, citrus pest categories, and pathway vector fruits (i.e., oranges, lemons, mandarins, grapefruits, and pomelos). These data were compiled from the Chinese National Pest Quarantine Information System. In accordance with the International Plant Protection Convention (IPPC), phytosanitary inspections of imported citrus species are generally conducted through risk-based probabilistic sampling, with the sampling intensity determined by factors such as the commodity type and interception history. They provide practical evidence for pathway-associated relative risks (Fenn-Moltu et al., 2023b; Turner et al., 2021), although they may still be influenced by trade exposure and surveillance intensity conditions. The 84 citrus pests intercepted at China's entry ports primarily consisted of scale insects (47), fruit flies (13), mites (3), and other pests (21) (Supplementary Table 1). Data on China's citrus import volumes from 2003 to 2018 were obtained from the UN Comtrade database (https://comtrade.un.org/). Citrus yield data derived from various provinces in China, indicating the sizes of the citrus industries in these provinces and spanning from 2013 to 2023, were sourced from the China Statistical Yearbook. The mean citrus production yield in each province was subsequently calculated.

2.2. Introduction risk analysis

Introduction risk networks were constructed for the citrus pests based on pest interception records obtained for five citrus species at Chinese entry ports. The interception records were log-transformed before performing the introduction risk analysis. We first assessed the introduction risks of citrus pests in different provinces of China on the basis of three key indicators: the number of intercepted pest species, the number of sources and the frequency of the interceptions. We then evaluated the robustness of the introduction risk networks based on two assessment strategies. The first strategy considered the importance levels of nodes and edges, where nodes represented sources and

entry ports, and edges represented the links between different sources and entry ports based on citrus pest interceptions; betweenness centrality (eq. 1) (Freeman, 1977), eigenvector centrality (eq. 2) (Bonacich, 2007), and the K-core algorithm were included as measures of node significance and the weights of edges, where betweenness centrality identified the key risk pathways, eigenvector centrality reflected the influences of connected risk nodes, and the K-core algorithm delineated the core structures sustaining introduction risks. The nodes and edges with greater importance levels were selectively removed, and a reanalysis was performed to assess the loss exhibited by the global efficiency (risk network robustness) of the introduction risk networks (Latora and Marchiori, 2001). The second strategy involved the random removal of risk nodes and edges, followed by an evaluation of the subsequent global efficiency reduction. On the basis of these calculations, we identified the optimal pest control strategies for mitigating the pest introduction risks associated with different citrus species.

$$C_B(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}} \quad (1)$$

$C_B(i)$ represents the betweenness centrality of risk node i. $\sigma_{st}(i)$ is the number of shortest paths from node s to node t that pass through node i. σ_{st} is the total number of shortest paths from node s to node t. A higher C_B value indicates that node i serves as a critical hub in the risk introduction network.

$$C_E(i) = \frac{1}{\lambda} \sum_j A_{ij} C_E(j) \quad (2)$$

$C_E(i)$ represents the eigenvector centrality of risk node i in the introduction risk network. A higher C_E value indicates greater importance for the risk of node i. A_{ij} is an element of the adjacency matrix of the network, and λ is the eigenvalue corresponding to the adjacency matrix.

We applied the 'fast unfolding' method to estimate the modularity structures of the introduction risk networks for citrus pests (Blondel et al., 2008). Here, modularity structures refer to network-based groupings of the source regions and entry ports in which the connections concerning pest introduction risks were denser within groups than between groups, reflecting distinct clusters of similar introduction risk profiles. For citrus pests with high interception frequencies, we calculated the potential introduction risk pathways associated with different citrus species, focusing specifically on the ten most frequently intercepted pest species. These pathways included fruits with specific introduction pathway vectors, source countries or regions, and entry ports.

2.3. Species distribution and environmental data

Species distribution data concerning citrus pests and environmental variables under the current and future climate conditions were used to support the ecological niche modeling process, with the aim of identifying potentially suitable areas for citrus pests. The global distribution records of various citrus pests were obtained from four sources: (1) the Global Biodiversity Information Facility (GBIF), (2) the Barcode of Life Data Systems database, (3) the Web of Science and China National Knowledge Infrastructure, and (4) our field survey data collected in China from 2021 to 2024. We performed an initial data-cleaning process to ensure the quality and accuracy of the distribution data. Distribution records were removed if they had clearly erroneous coordinates (e.g., locations in the sea or coordinates with identical latitudes and longitudes). Additionally, to reduce potential sampling and spatial biases and to alleviate the influences of spatial autocorrelations on the model calibration and evaluation processes, we used the ENMTools package to retain only one distribution record for each citrus pest within each grid cell of the environmental variables (Warren et al., 2021). Consequently, for 84 citrus pests, we removed 23 citrus pests whose distribution record counts were less than 20. We retained 61 citrus pests for modeling their potentially suitable habitats, including 27 scale insects, 12 fruit flies, and 22 other pests (Supplementary Table 1).

We obtained current climate data (1981–2010) and future climate projections (2011–2040) at a 1-km resolution from the CHELSA V2.1 database, which included 19 bioclimatic variables (Karger et al., 2017). Future climate data were collected from three shared socioeconomic pathways (SSPs): SSP126, SSP370, and SSP585. To minimize the uncertainty derived from the use of a single generalized circulation model (GCM), we selected four GCMs (GFDL-ESM4, MPI-ESM1–2-HR, MRI-ESM2–0, and UKESM1–0-LL) and averaged their outputs (Karger et al., 2021). The elevation data produced at a 5-km resolution were collected from the WorldClim database. We also collected human footprint (HFP) data for 2018 from the global human footprint maps developed by Mu et al. (2022). The human footprint index ranged from 0 to 50, with higher values indicating greater human impacts on the environment. We used 2021 net primary production (NPP) data at a spatial resolution of 500 m, which were obtained from the MOD17A3H Version 6 data product, as an indicator of the availability of resources and the intensity of species interactions. We included these environmental variables because the citrus pest establishment and spreading processes are jointly shaped by climate suitability, landscape characteristics, and anthropogenic activities. All environmental layers, including climate variables, elevation, the human footprint, and the NPP, were resampled to a uniform 5-km resolution to ensure consistency across different datasets. To minimize the potential impact of multicollinearity among the environmental variables on the performance of the model, we conducted Pearson's correlation analyses for all the variables and, when highly correlated variables were identified, preferentially retained the predictors with clearer biological relevance to the growth, development, survival, and establishment of citrus pests (Liu et al., 2019a; Petitpierre et al., 2017). The final set of environmental variables with low correlation coefficients ($|r| < 0.7$) was retained for ecological niche modeling purposes (Supplementary Table 2).

2.4. Establishment and spread risk analysis

The establishment and spread risks of citrus pests in China were identified using an ensemble ecological niche modeling approach, which effectively minimized the uncertainty associated with predictions derived from a single ENM (Araújo and New, 2007). This approach has been widely applied in species distribution modeling scenarios (Liu et al., 2019a; Moura et al., 2023; Pacifici et al., 2019). In this study, we employed five algorithms that are commonly used in machine learning-based ENMs: the generalized linear model (GLM), a random forest (RF) (Breiman, 2001), a gradient-boosting machine (GBM), the maximum entropy model (MaxEnt) (Phillips et al., 2017), and multivariate adaptive regression splines (MARS). These models were used to characterize species–environment relationships and generate habitat suitability predictions, and they have demonstrated a strong ability to predict the distributions of invasive species and pests (Liu et al., 2019b; Moura et al., 2023). Before conducting model training, we defined a 200-km buffer around each pest distribution record (Liu et al., 2019b; Van Der Wal et al., 2009). Within this buffer, pseudoabsence points were randomly generated at a ratio of 0.5 (presence-absence) (Moura et al., 2023). We employed 5-fold cross-validation. Specifically, 80% of the distribution data were used for model calibration, and the remaining 20% were used for model evaluation purposes (Roberts et al., 2017). We evaluated the performance of the species distribution models using two commonly applied metrics, namely, the area under the receiver operating characteristic curve (AUC) and the true skill statistic (TSS), with the AUC reflecting the overall discrimination ability of each model and the TSS indicating its classification accuracy (Leroy et al., 2018; Liu et al., 2011). Higher values of both metrics indicate better model performance. On the basis of the model evaluation results, individual models with TSS values greater than 0.6 and AUC values

exceeding 0.7 were selected for ensemble modeling with the weighted mean method on the biomod2 platform (Thuiller et al., 2009). To produce binary suitability maps for each citrus pest, the continuous outputs of each ensemble model were converted into suitable and unsuitable areas using the threshold that maximized the TSS value. The potentially suitable areas for each citrus pest in China under both the current and future climate conditions were then calculated to assess its establishment risk. We further performed a spatial overlay analysis of the suitable habitat maps for 61 citrus pests to estimate the species richness level, which was used as an indicator of the establishment risk across different regions of China. In addition, we separately identified the spatial species richness patterns of two major pest groups, namely, 27 scale insects and 12 fruit flies, to determine their establishment risks in China. Finally, we quantified the projected species richness changes under future climate conditions to identify potential high-risk pest spread regions. We also calculated the mean species richness for citrus pests at the provincial level under both the current and future climate conditions, as well as the richness changes observed between the two periods, to clarify the establishment and potential spread risks of the citrus pests contained in each province of China.

2.5. Risk management system

To support multiple-risk-node management for the introduction, establishment and spread of citrus pests along trade pathways, we developed an integrated decision support platform that couples a large language model (Qianwen-3 Plus) with an RAG mechanism (Lewis et al., 2021; Li et al., 2025). The system was designed to translate quantitative risk assessment results into actionable management guidance for regulators and quarantine authorities.

The system integrates heterogeneous information sources into its retrieval corpus, including (i) citrus pest interception records and derived introduction risk network outputs, (ii) the establishment of suitability outputs and provincial species richness summaries, (iii) the projected spread risk results obtained under future climate scenarios, and (iv) citrus importing/trade and provincial production statistics. Users can interact with the system through natural-language queries (e.g., "*Which citrus pests pose the highest introduction risks through lemon imports*?" or "*Which provinces face the highest establishment and spread risks under future climate scenarios*?"). To enable coherent reasoning across multiple risk stages, all the quantitative analysis results and relevant documents were transformed into a unified knowledge base. This knowledge base supports the retrieval of risk information across the introduction, establishment, and spread stages of citrus pests. The RAG framework was employed to ensure that the responses were grounded in verified data rather than relying solely on the internal knowledge of the language model.

The decision support platform was implemented using a modular and containerized architecture to ensure its scalability and reproducibility. The front end was developed with Vue3, while the Java-based RESTful backend handles data processing and inference requests. Relational data (e.g., system metadata) are stored in MySQL, Redis is used for in-memory caching, and MongoDB manages the intermediate document processing results. A PostgreSQL database with the pgvector extension serves as the core vector knowledge base, enabling efficient high-dimensional similarity searches. Heterogeneous data sources (e.g., text documents, tables, and quantitative risk outputs) were converted into dense semantic vectors using the text-embedding-v3 model (OpenAI, 2024) and stored in the vector database. During the inference process, both dense (vector-based) and sparse (keyword-based) retrieval are performed, and the results are combined using reciprocal rank fusion and reranked with the BGE-reranker-v2 model (BAAI, 2024) to improve their relevance. The retrieved evidence is injected as contextual grounding information into the Qianwen-3 Plus large language model, ensuring that the generated risk analyses and management suggestions remain factually grounded and contextually coherent. To further enhance the analytical robustness of the system, it supports a deep reasoning mode that enables extended contextual inference and real-time online information retrieval processes to complement the existing knowledge base (e.g., with pest interception frequency and trade volume information) (Li et al., 2025).

Because the platform relies on a curated domain-specific retrieval corpus, the purpose of this study was to evaluate the performance of the RAG-grounded system itself rather than to benchmark it against a generic LLM without access to the same evidence base. The performance of the decision support system was evaluated using 30 question–answer pairs representing typical multistage citrus pest risk management scenarios. The similarity between the system-generated answers and reference answers was assessed using three methods: cosine similarity, BM25, and BERT (Devlin et al., 2019; Robertson and Zaragoza, 2009). These methods were used to assess the ability of the platform to retrieve, interpret, and rank relevant knowledge derived from the vector database with respect to the input queries.

3. Results

3.1. Introduction risks in different provinces of China

The spatial distribution of the citrus pest introduction risks observed across different Chinese provinces indicated notable regional variations, with distinct patterns emerging for each citrus species considered an introduction vector (Fig. 2). On the basis of the interception data produced across various citrus species, an increasing trend in the risk of citrus pest introduction in China was observed from 2003 to 2018 (Supplementary Fig. 1). Moreover, China's citrus import volumes tended to increase overall from 2003 to 2018 across all the citrus species (Supplementary Fig. 2). GD consistently presented the highest risk of citrus pest introduction, which was characterized by the highest interception frequency, number of source countries or regions, and number of intercepted species, including overall citrus species (Fig. 2a), orange (Fig. 2b), lemons (Fig. 2c), mandarin (Fig. 2d), grapefruit (Fig. 2e), and pomelo (Fig. 2f). SH, SZ, and FJ were frequently identified as regions associated with high secondary introduction risks. Notably, YN showed an elevated risk specifically for pest introductions via lemons, highlighting its unique role among provinces with high introduction risks (Fig. 2c). In contrast, SN, HB, BJ, and HL consistently presented low risks regarding the introduction of citrus pests with various citrus species.

3.2. Introduction risk networks

We systematically identified the key nodes with strong betweenness and eigenvector centrality in the introduction risk networks for the citrus pests mediated by different citrus species (Fig. 3a-f) (Supplementary Tables 4–9). Overall, the Chinese provinces GD, SH and SZ showed high betweenness and eigenvector centrality levels for the introduction risk of entry ports (Supplementary Tables 4–9). Our results also indicated that the key risk nodes displayed certain common structural characteristics in the subnetworks for the introduction of citrus pests via different citrus species. The provinces GD, SH, and SZ consistently occupied central positions in the introduction risk networks for citrus pests via oranges, mandarins, and grapefruits, respectively. However, province FJ demonstrated high betweenness and eigenvector centrality in the introduction risk networks for citrus pests via lemon and pomelo. The source countries or regions associated with citrus pest introduction risks varied substantially among the different citrus species (Fig. 3a–f). Specifically, certain source countries/regions (e.g., TW and THA) consistently showed high centrality across multiple introduction risk networks, indicating a broad potential for pest introduction, whereas others (e.g., ZAF, USA, and SGP) played pivotal roles only within specific introduction risk networks (Supplementary Tables 4–9). TW and THA exhibited high rankings in terms of both betweenness and eigenvector centrality within the citrus pest

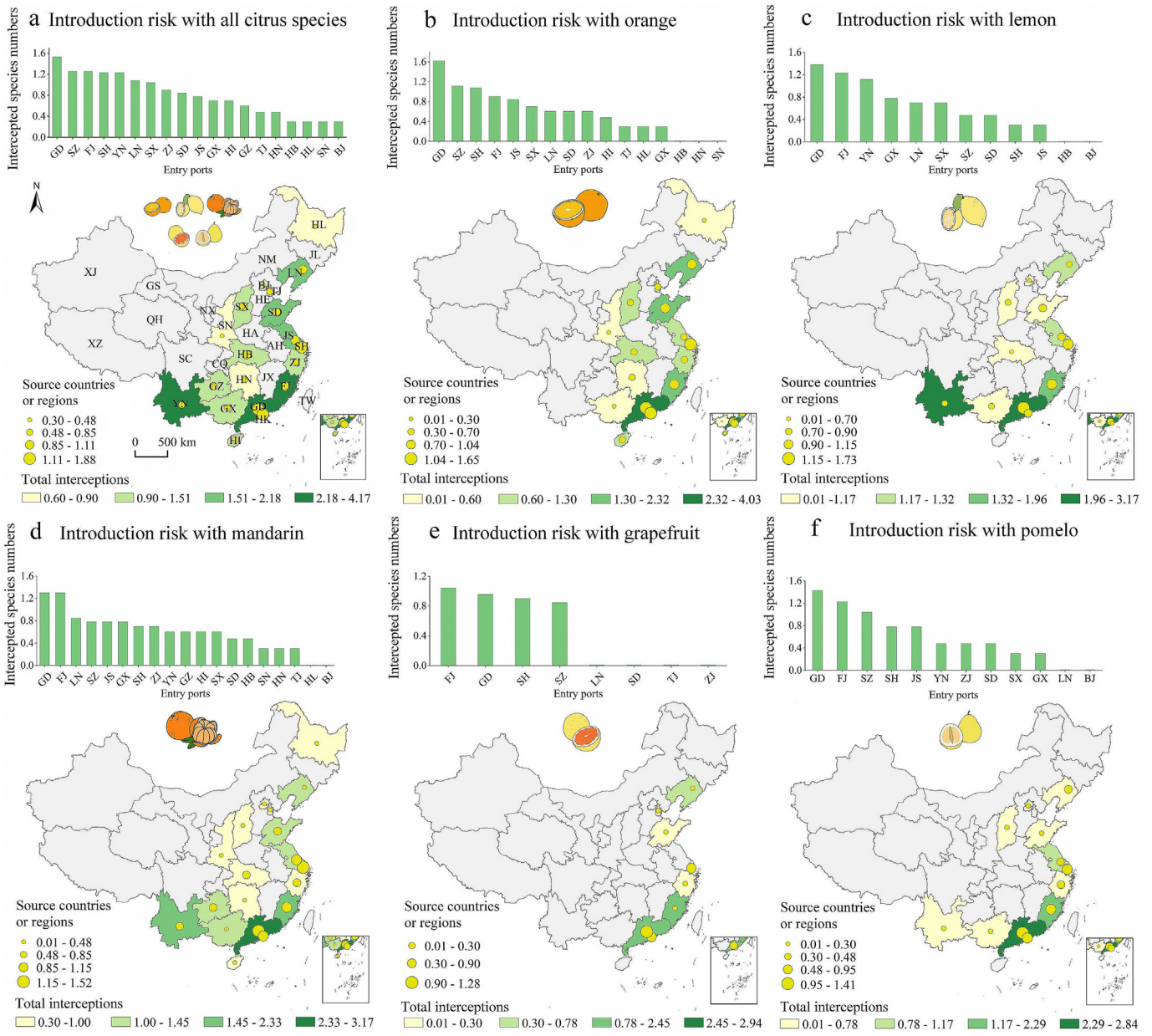


Fig. 2. Spatial patterns of citrus pest introduction risks delineated via (a) the overall citrus species, (b) oranges, (c) lemons, (d) mandarins, (e) grapefruits, and (f) pomelos across different Chinese provinces, quantified using the total number of pest interceptions, the number of intercepted species, and the number of source countries or regions. Note: The abbreviations for the examined provinces are given using two letters of their official names; detailed explanations can be found in Supplementary Table 3.

introduction risk networks for the overall citrus species, mandarins, and pomelos. SGP and HK demonstrated high betweenness and eigenvector centrality in their introduction risk networks of citrus pests for oranges. The USA and VNM showed notable betweenness centrality, whereas HK and TW exhibited greater eigenvector centrality in the introduction risk network of citrus pests for lemons. ZAF and EGY occupied prominent positions in the citrus pest introduction risk network for grapefruits.

We evaluated the robustness of citrus pest introduction risk networks using global efficiency metrics and found that their structural integrity strongly depends on specific risk nodes. The results revealed that the removal of key risk nodes (e.g., provinces/countries and regions GD, SH, SZ, TW, and THA) on the basis of betweenness and eigenvector centrality, as well as the k score, led to a rapid and significant global efficiency decline in all the citrus pest introduction risk networks (Fig. 3g-l), indicating that nodes with high centrality and k-core importance levels play key roles in the introduction risk networks of citrus pests. In contrast, the random removal strategy required the removal of more risk nodes to achieve a similar global efficiency loss. Prioritizing quarantine and biosecurity measures at these key source regions and entry ports could therefore substantially increase the effectiveness of risk management strategies for citrus pests.

Our results revealed 243 introduction risk pathways and 98 risk nodes, forming five major modularity classes in the citrus pest introduction risk network that was mediated by all the citrus species (Fig. 4a). The orange-mediated subnetwork comprised 162 introduction risk pathways among 69 risk nodes, forming 5 modularity classes (Fig. 4b). The lemon-mediated subnetwork included 118 introduction risk pathways and 71 risk nodes, forming four modularity classes (Fig. 4c), whereas the mandarin subnetwork included 130 introduction risk pathways and 67 risk nodes, forming five modularity classes (Fig. 4d). In contrast, the grapefruit-mediated subnetwork was notably sparse, with 41 introduction risk pathways and 30 risk nodes (Fig. 4e), whereas

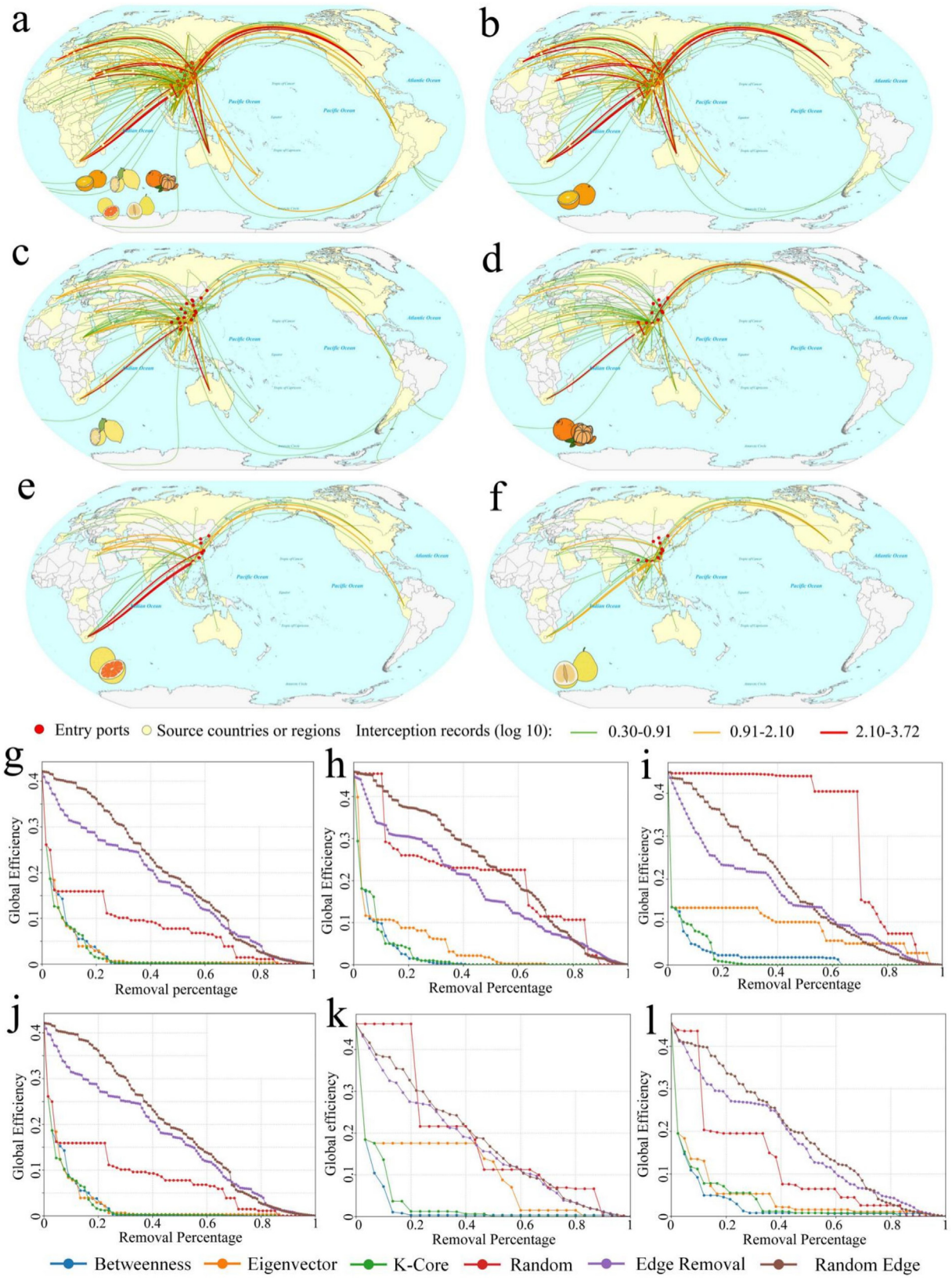


Fig. 3. Spatial configurations and robustness levels of the citrus pest introduction risk networks produced across various citrus species. (a–f) Risk networks constructed for (a) all the citrus species, (b) oranges, (c) lemons, (d) mandarins, (e) grapefruits, and (f) pomelos. Each network comprises source countries or regions and entry ports as risk nodes, with edge weights representing the associated interception frequencies. Red dots denote entry ports; yellow circles indicate the source countries or regions. (g–l) Robustness of the citrus pest introduction risk networks mediated by (g) all citrus species, (h) oranges, (i) lemons, (j) mandarins, (k) grapefruits, and (l) pomelos, evaluated through the global efficiency index (robustness of the introduction risk networks) under two disruption scenarios: (1) the targeted removal of critical nodes and edges based on betweenness centrality, eigenvector centrality, and the k-core algorithm, and (2) the random removal of nodes and edges. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

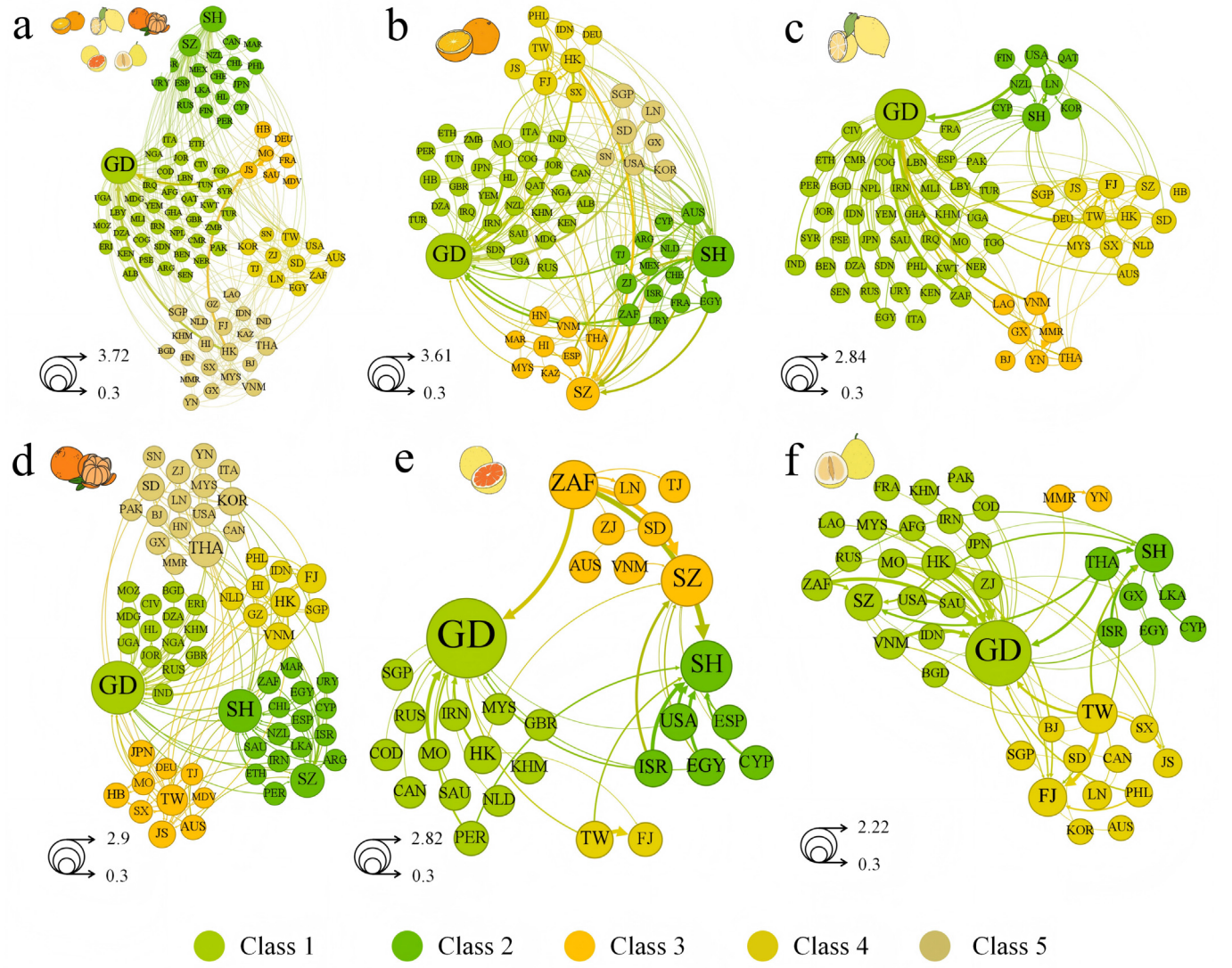


Fig. 4. Modularity classes of the citrus pest introduction risk networks mediated by different citrus species: network-based groupings of the source countries or regions and entry ports with dense pest introduction risk connections. The network structures represent the modular organization schemes of the citrus pest introductions mediated by (a) all citrus species, (b) oranges, (c) lemons, (d) mandarins, (e) grapefruits, and (f) pomelos. Each node denotes a risk source or entry port, and the size of a node corresponds to its connectivity within the network. The colors indicate the modularity classes (Classes 1–5), reflecting the clusters of regions with closely associated citrus pest introduction pathways. Note: The abbreviations for the examined countries follow the ISO 3166-1 alpha-3 standard, and the abbreviations for regions including Hong Kong, China; Macao, China and Taiwan, China are HK, MO and TW, respectively (see Supplementary Table 3).

the pomelo-mediated subnetwork exhibited 69 introduction risk pathways and 42 risk nodes (Fig. 4f); both networks consisted of four modularity classes. In addition, our results indicated that the introduction risk networks mediated by all the citrus species, namely, oranges, lemons, and mandarins, were highly connected, whereas those mediated by grapefruits and pomelos exhibited markedly lower connectivity and modular complexity levels.

The introduction risk pathways of citrus species mediated by various citrus pests, mainly *Aonidiella aurantii*, *Parlatoria zizyphus*, and *Lepidosaphes beckii*, followed by *Chrysomphalus ficus* and *Unaspis yanonensis*, were associated with the greatest introduction risks for scale insects (Fig. 5). Specifically, the primary introduction risk pathways for *A. aurantii* were introductions from HK to GD and SZ, from the USA to GD and SH, and from ZAF to SH and SZ, which was mediated mainly by oranges, lemons, pomelos, and mandarins (Fig. 5a-f). The primary introduction risk pathways for *P. zizyphus* were introductions from MO and HK to GD, which were mediated mainly by oranges and mandarins (Fig. 5b and d, respectively). The primary introduction risk pathways for *L. beckii* were introductions from EGY to SH and SZ and from ZAF and the USA to GD, which were mediated mainly by oranges and lemons (Fig. 5b and c, respectively). Fruit flies also presented a high introduction risk, particularly in *Bactrocera dorsalis*. The primary introduction risk pathways for *B. dorsalis* were introductions from HK and MO to GD, from TW to FJ, and from VNM to GX, which were mainly mediated by oranges and mandarins (Fig. 5b and d, respectively). Potentially important invasive species, namely, *Cryptophlebia leucotreta*, also presented high introduction risks. The primary introduction risk pathways for *C. leucotreta* were introductions from ZAF to GD and SH, which were mediated mainly by oranges and grapefruits.

3.3. Establishment and spread risks

The mean TSS and AUC values of the ensemble modeling results were 0.88 and 0.97, respectively, indicating that the ensemble model for citrus pests was highly rigorous (Supplementary Fig. 3). The annual mean temperature (bio1) and human footprint (HFP) were the high-

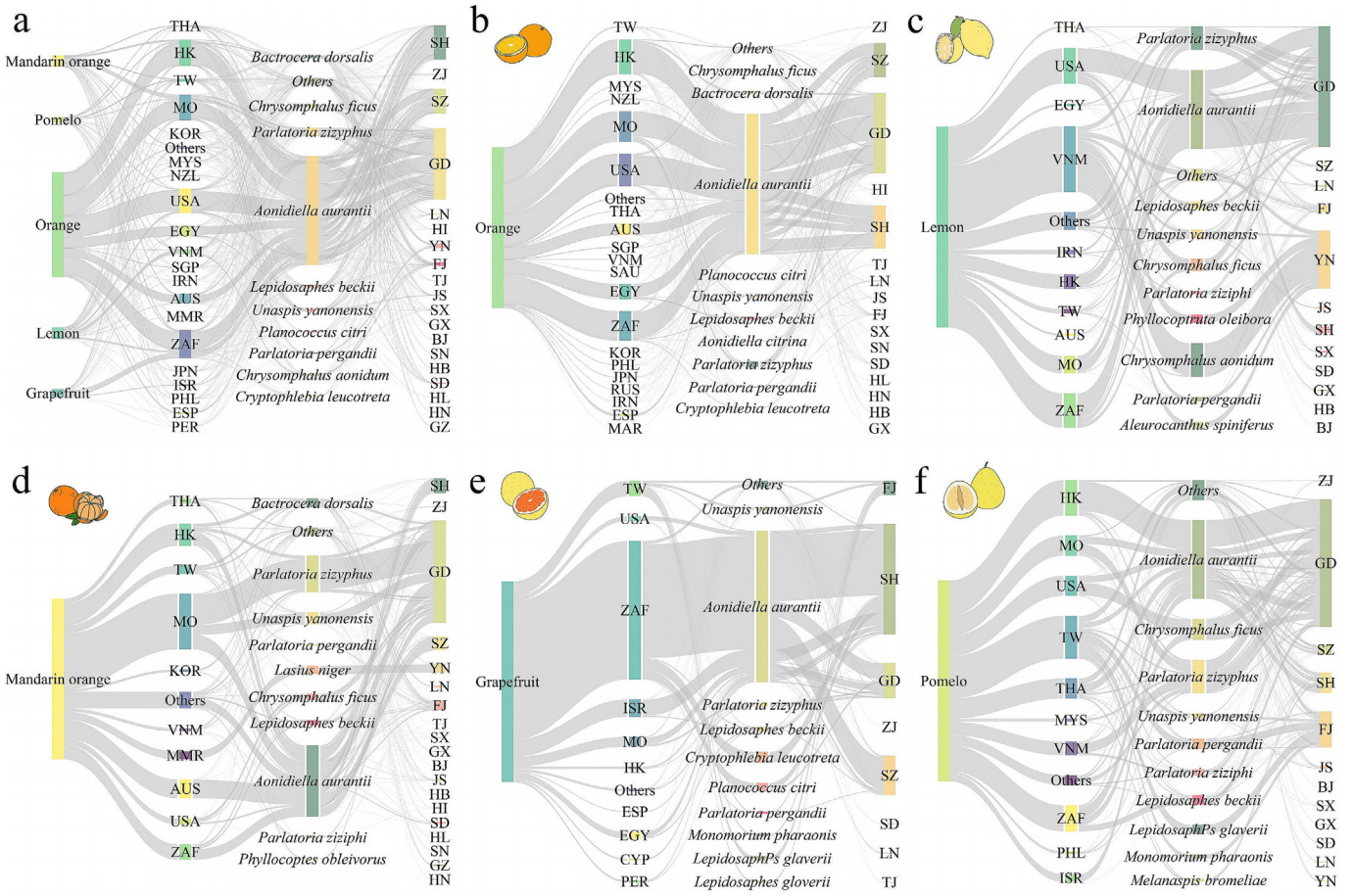


Fig. 5. Pathways of the introduction risks for the high-risk citrus pests mediated by different citrus species, which illustrate the multilevel introduction risk structures from specific citrus species to the source risk countries or regions, followed by the target citrus pest, and finally to the potential entry ports in China. The panels represent the pathways mediated by (a) all citrus species, (b) oranges, (c) lemons, (d) mandarins, (e) grapefruits, and (f) pomelos.

ranking variables that affected the potentially suitable areas for citrus pests (Supplementary Fig. 4). Our results indicated that several citrus pests, including *Bactrocera scutellata* (249.07×10^4 km^2), *Pseudococcus comstocki* (222.94×10^4 km^2), and *Chrysomphalus aonidum* (213.12×10^4 km^2), are potentially extensively distributed under the current climate conditions in China (Supplementary Table 10). Other citrus pests, such as *Aleurocanthus spiniferus* and *Bactrocera tau*, also maintained largely potentially suitable areas, exceeding 190×10^4 km^2 under the current climate conditions. Most citrus pests with high establishment risks are projected to further expand their potentially suitable areas under future climate scenarios. The potentially suitable areas of *Bactrocera scutellata*, *P. comstocki*, and *C. aonidum* were expected to increase to 271.28×10^4 km^2, 244.10×10^4 km^2, and 227.42×10^4 km^2, respectively, under the SSP585 scenario. Similarly, the potentially suitable areas for *B. dorsalis* and *B. zonata* increased markedly, from 114.89×10^4 km^2 to 139.24×10^4 km^2 and from 88.73×10^4 km^2 to 125.08×10^4 km^2, respectively, under the SSP585 scenario. Conversely, the potential suitable areas for several citrus pests, such as *Aonidiella aurantii*, *Ceratitis capitata*, and *Grapholita molesta*, decreased under future climate scenarios.

Citrus production in China is concentrated in the southern and southeastern provinces, particularly GX (1029.81×10^4 t), HN (544.30×10^4 t), GD (490.33×10^4 t), HB (483.02×10^4 t), and SC (452.83×10^4 t) (Fig. 6a). These provinces presented the highest mean citrus pest species richness, particularly SH (45.92), JS (33.13), GD (30.72), HN (28.60), and GX (27.27), under the current climate conditions, indicating the overlap between high-yield citrus provinces and the establishment risk hotspots for citrus pests (Fig. 6b-d) (Supplementary Table 11). In contrast, the northern and western provinces in China (e.g., GS, NX, XJ, and QH) presented low species richness levels for citrus pests (<3) (Fig. 6b-d) (Supplementary Table 11). The mean species richness and establishment risks of scale insects were high in SH (18.87), GD (13.02), HN (11.32), and GX (11.51) (Fig. 6c), whereas those for fruit flies were greatest in SH (7.98), YN (5.17), GZ (6.03), GX (6.77), GD (7.14), and HI (6.82) under the current climate conditions (Fig. 6d and Supplementary Table 11).

The overall spatial pattern of the citrus pest richness levels remained relatively stable but showed distinct regional trends under future climate scenarios (Fig. 6e–g) (Supplementary Figs. 5 and 6 and Supplementary Table 12). The southern provinces in China, including GD, GX, and HI, showed gradual citrus pest richness reductions under future climate scenarios. In contrast, the provinces in the Yangtze River Basin, such as HN, SC, and HB, showed modest increases in their citrus pest richness, indicating a potential northward shift and spread in the establishment risk of citrus pests.

3.4. Risk management framework and system

We developed a multistage risk management framework for plant pests in the food trade, integrating introduction risk prevention, establishment risk surveillance, and spread risk containment (Fig. 7). The introduction stage focuses on identifying potential risk pathways and nodes, thereby prioritizing high-risk vectors and plant pests. In the establishment stage, the climate suitability and risk zonation of plant

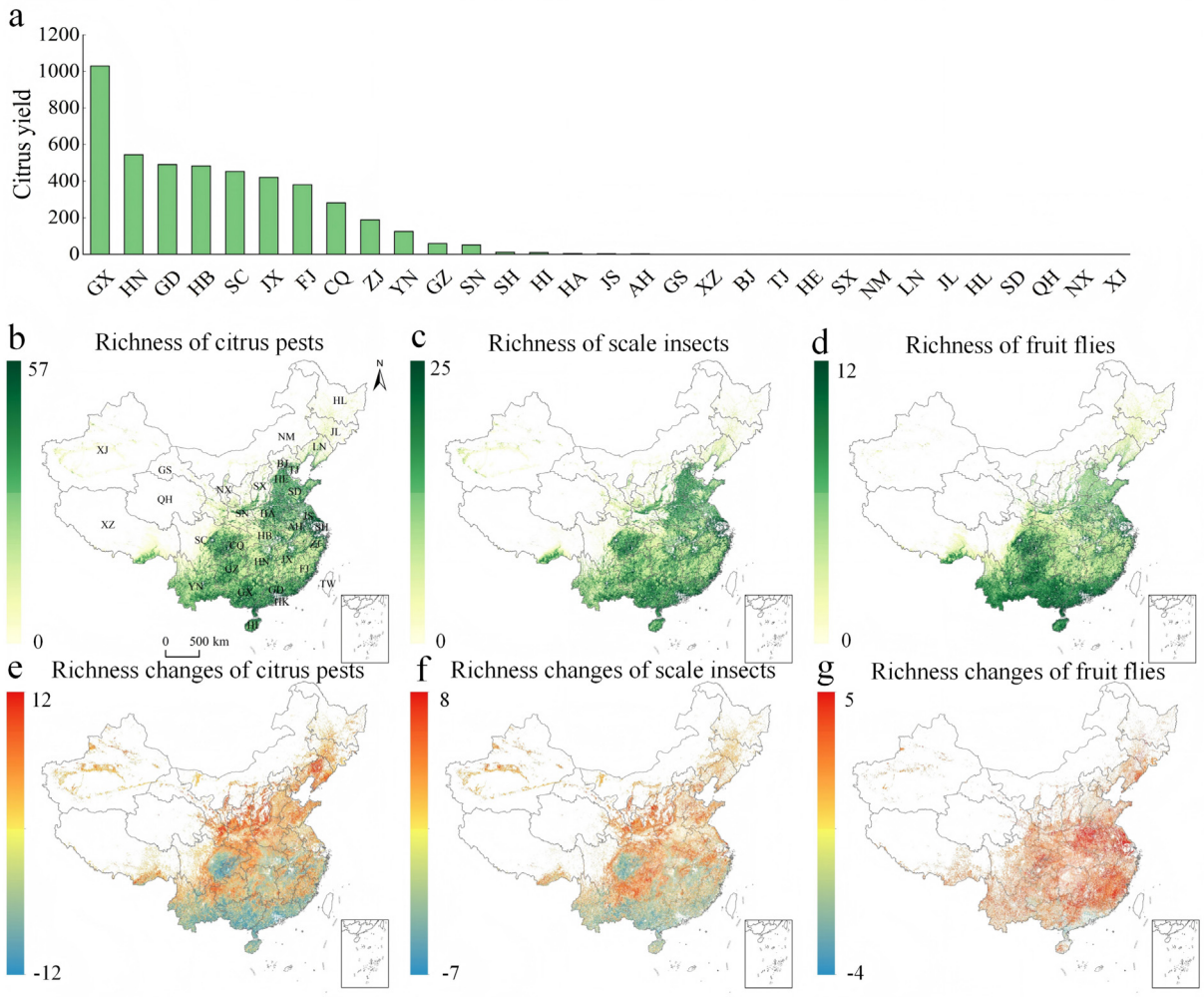


Fig. 6. Citrus yields and spatial patterns of the citrus pest richness levels observed under current and future climate scenarios in China. (a) Citrus yield ($\times 10^4$ tons) across the citrus-producing provinces in China. (b–d) Species richness of citrus pests (b), scale insects (c), and fruit flies (d) under the current climate conditions. (e–g) Projected changes in the species richness of citrus pests (e), scale insects (f), and fruit flies (g) under the SSP126 climate scenario.

pests are estimated, whereas in the spread stage, the spatial dynamics of risks and regional containment strategies are addressed. By linking these three stages, the framework supports a more holistic pest management approach that connects preborder prevention, domestic surveillance, and postestablishment containment across the full process, improving the prioritization of monitoring and intervention efforts. We thereafter developed a decision support system for managing the risks of plant pests by coupling a large language model

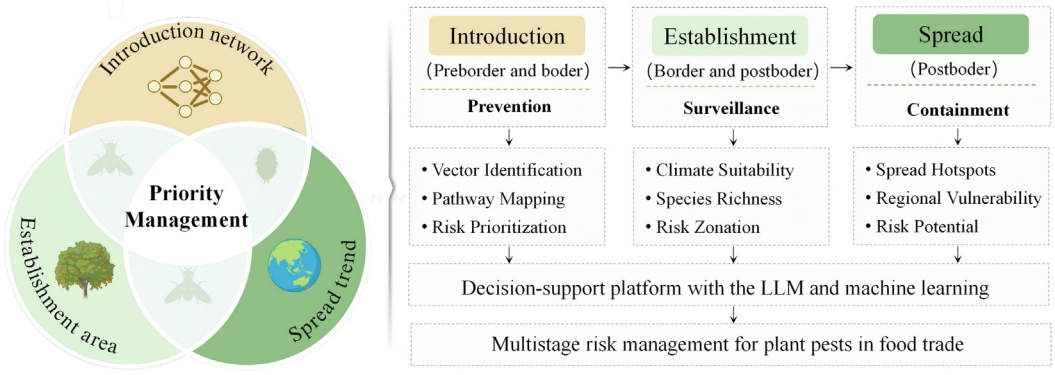


Fig. 7. Multistage risk management framework for plant pests in the food trade. The framework integrates introduction risk prevention (i.e., vector identification, pathway mapping, and risk prioritization), establishment risk surveillance (i.e., climate suitability, species richness, and risk zonation), and spread risk containment strategies (i.e., spread hotspots, regional vulnerabilities, and risk potential). On the basis of this framework, a decision support platform for the management of plant pest risks in the food trade was developed by integrating a large language model and machine learning.

(Qianwen-3 Plus) with an RAG mechanism (Supplementary Fig. 7). The performance test results of the decision support platform for the risk management of plant pests in the food trade based on 30 question–answer pairs related to multistage risk management for citrus pests revealed that the mean accuracy scores produced for Cosine, BM25, and BERT were 0.76, 0.9648, and 0.8626, respectively, indicating its ability to reliably match relevant knowledge with complex risk management schemes for plant pests in the food trade (Supplementary Table 13).

4. Discussion

Managing the risks posed by plant pests in the global food trade has emerged as a critical priority because of increasing international trade and climate change (Ali et al., 2023; Ma et al., 2025; van Klinken et al., 2023). Compared with the previously proposed pest risk frameworks that typically focus on a single risk stage, a specific pest group, or a single modeling technique, we developed a multistage risk management framework with multiple food trade pathways, including the introduction, establishment, and spread of pests. Additionally, our study integrated a network-based introduction risk analysis, machine learning-based ecological niche modeling, and a retrieval-augmented large language model through an AI-assisted decision support platform to systematically assess and manage the citrus pest risks (i.e., introduction, establishment, and spread risks) that are associated with the citrus trade. Importantly, the LLM is an evidence-grounded decision interface that is supported by a RAG pipeline; it retrieves structured outputs from quantitative models and associated documents. This design could effectively improve the accessibility and interpretability of multistage pest risk information for regulatory decision-making processes and translate this risk information into actionable management guidance, providing reliable insights for identifying, evaluating, and mitigating pest risks in the food trade.

The identification of high-risk plant pests, their introduction risk pathways, their source countries or regions, and the entry ports that are mediated by different food trade pathway vector commodities is crucial for prioritizing prevention strategies for high-risk commodity pests in the global food trade (Capinha et al., 2023; Fenn-Moltu et al., 2023a; Nardi et al., 2025). Citrus is among the most important commodities in the international fruit trade and represents a major imported fruit category in China. Our findings revealed a continuous increase in the introduction risk of citrus pests in China's citrus imports since 2003, as reflected in both the number of intercepted pest species and their interception frequency. This increase coincided with steady growth in the import volumes of citrus, indicating that higher import volumes were likely associated with more frequent pest interceptions. Previous studies revealed that global trade, such as that involving the importing of fruits and crop products, facilitated the introduction risk of plant pests (Bertelsmeier et al., 2025; Bonnamour et al., 2021; Chapman et al., 2017). Since China's accession to the World Trade Organization (WTO), citrus imports from major trade partners, particularly ASEAN countries and regions along the Belt and Road, have substantially increased, which may have heightened the likelihood of pest introductions through multiple citrus trade pathways (Gu et al., 2021; Zhao et al., 2024b). Our findings are the first to identify the specific source countries or regions and entry ports that are associated with citrus pest introductions mediated by various citrus species. GD, SH, SZ, YN, and FJ were consistently identified as high-risk entry ports in the introduction risk network of citrus pests. Provinces GD, SZ, and SH were exposed to multiple source countries or regions and citrus pests, indicating the importance of comprehensive and pathway-specific quarantine strategies that target high-risk pests and trade pathways. In contrast, YN and FJ faced introduction risks that were dominated by multiple citrus pests introduced from specific source countries or regions, suggesting that targeted quarantine measures focused on high-risk sources were effective. ZAF, the USA, EGY, VNM, TW, MO, and HK emerged as high-risk sources characterized by both high pest diversity levels and high interception frequencies. Notably, these sources also represented major citrus trade partners of China, further underscoring the trade volume as a key determinant of citrus pest introduction risks. Our findings highlight the importance of strengthening the development of quarantine and early warning strategies at high-risk sources and entry ports.

Our findings indicate that the risk sources and entry ports of the citrus pest introductions mediated by different citrus species were heterogeneous. For instance, YN was identified as a specific high-risk entry port within the lemon-mediated introduction network of citrus pests, whereas the USA and VNM served as high-risk sources. These findings provide a theoretical basis for prioritizing risk sources and entry ports when managing the citrus pest introduction networks that are associated with distinct citrus species. Scale insects and fruit flies represent the categories with the highest introduction risks. Previous studies have documented the widespread dispersal of scale insects and fruit flies across the global fruit trade (Mani and Shivaraju, 2016; Zhao et al., 2024c). Our results further revealed that multiple scale insects (e.g., *A. aurantii* and *P. zizyphus*) and fruit flies (e.g., *B. dorsalis*) presented high risks of cointroduction via different citrus hosts at China's entry ports, imposing high pressure on management personnel during preborder prevention and border quarantine processes. We identified other high-risk potential invasive species that are not yet in China, such as *C. leucotreta* and *C. capitata*, and estimated their primary pathway vectors and pathways of introduction. They have caused very large economic losses in the production of citrus and other fruit crops worldwide and represent a persistent potential threat (Githae et al., 2025; Hallouti et al., 2024). Our findings can support the development of targeted prevention strategies for citrus pests based on their specific risk pathway vectors and pathways, especially for potentially invasive species.

The estimation of potentially suitable areas and species richness levels for different plant pests across crop planting regions, as well as their spread risk dynamics under future climate scenarios, provides new insights into risk zonation and classification, as well as surveillance and containment schemes (Bebber et al., 2013; Savary et al., 2019; Wang et al., 2022). Our findings indicated that scale insects (e.g., *P. comstocki* and *C. aonidum*) and fruit flies (e.g., *B. scutellata*, *B. dorsalis*, and *B. zonata*) presented high establishment risks in the important citrus planting regions of China, particularly in southern and Yangtze River Basin provinces such as Guangxi, Hunan, Guangdong, Hubei, and Sichuan. Specifically, these regions face the compounded threat posed by the co-occurrence of multiple scale insects and fruit flies. Scale insects and fruit flies are projected to shift their suitable ranges poleward under future climate scenarios, increasing their establishment risks at higher latitudes. Specifically, scale insects posed a high spread risk in the Yangtze River Basin provinces. Fruit flies exhibited greater overall spread risk potential, with a high spread risk across the citrus planting regions of the southern and Yangtze River areas. Previous studies have also indicated that climate change facilitates the expansion of the ranges scale insects and fruit flies in tropical and subtropical areas while simultaneously increasing their spread risks in higher-latitude regions (Gutierrez et al., 2021; Zhao et al., 2024a). By identifying the species- and region-specific establishment and spread risks associated with scale insects and fruit flies, our findings provide actionable guidance for prioritizing targeted surveillance and containment measures in China's citrus planting regions.

Integrating the introduction, establishment, and spread risks associated with citrus pests in China indicated that the citrus planting regions located in southern China (e.g., Guangdong, Guangxi, and Fujian) face multistage and compound threats from the co-occurrence of multiple citrus pests. To support targeted prevention, surveillance, and containment measures, we proposed a multistage risk management framework and developed a decision support system that combines quantitative risk analysis data and a result knowledge vector database for citrus pests. Generative AI, including large language models, has seen growing

applications as a decision support tool in plant pest risk management scenarios (Ali and Zhuo, 2025; Rafiq et al., 2025; Yan et al., 2025; Zhang et al., 2024). However, their decision support effectiveness is still limited by the quality and timeliness of their underlying training data. Our findings indicate that the decision support system provides reliable decision-making guidance for trade-related plant pest risk management tasks by linking this domain-specific knowledge vector database to the developed LLM using the RAG approach. The RAG-enabled decision support system for plant pest risk management achieved a peak accuracy exceeding 0.9 in prototype Q&A evaluations based on the knowledge vector database of multistage citrus pest risks. The results of the present evaluation indicate that the system can reliably connect structured quantitative risk outputs with the LLM-based response generation process, thereby providing a practical reference for managers and regulators to complete data-driven risk management for plant pests in the food trade. Further studies should focus on enhancing the dynamic updating capacity of the underlying knowledge base and improving its alignment with retrieval strategies, along with a more rigorous system validation. In addition, integrating complementary techniques, such as prompt engineering and dedicated knowledge graph construction, could further increase the reliability and practical decision support capacity of the system (Chen et al., 2021; Ding et al., 2024; Mol and Kumar, 2024).

This study demonstrates that integrating multistage risk assessments with AI-assisted decision support can provide guidance for the development of decisions and multistage risk management strategies for food trade-related plant pests. For policymakers and regulatory authorities, the proposed framework offers a concrete basis for prioritizing quarantine inspections, allocating surveillance resources, and designing targeted biosecurity interventions at high-risk sources, entry ports, and production regions. Moreover, the RAG-enabled decision support system enables regulators to access up-to-date, evidence-based risk information through natural-language interactions, thereby improving the transparency and efficiency of plant pest risk management decisions. Compared with computationally intensive fine-tuning strategies, the RAG-based integration scheme may be particularly well suited to plant biosecurity applications, where decision contexts are inherently multistage, taxon-specific, and data-heterogeneous considerations and where risk evidence often requires flexible updating procedures in response to newly available interception records, distribution data, and environmental change projections. Overall, this approach provides a scalable and transferable model for supporting science-based policy-making work when managing trade-related plant pest risks under increasing globalization and climate change (Ali et al., 2023).

This study has several limitations. The introduction risk estimates derived from interception records and trade-related information may have been affected by factors such as trade volumes, inspection intensities, and reporting practices and therefore should be interpreted as relative indicators of pathway-associated relative risks under a risk-based phytosanitary inspection system. The establishment and spread risk analyses primarily reflected environmental suitability and projected risk patterns; as such, they may not fully represent invasion dynamics under the influences of management interventions or dispersal barriers. The LLM-RAG decision support module was evaluated using a domain-specific citrus pest case study, providing an initial demonstration of the practical utility of the framework. Importantly, the validation was based on 30 QA pairs and represents a task-specific functional assessment of the system's question-answering consistency and decision-support capability for the multistage risk management scenario of citrus trade pests. A broader validation across additional pest groups, commodity systems, and regulatory contexts would further strengthen its generalizability. Future work should incorporate expert-annotated benchmark datasets and multidimensional evaluation protocols, including expert-based validation, spatial block validation, uncertainty propagation analysis, and benchmarking against simpler or conventional decision-support systems, to more rigorously assess the reliability, robustness, and practical decision-making ability of the system across broader pest groups and regulatory scenarios.

CRediT authorship contribution statement

Haoxiang Zhao: Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Shanqing Yi:** Writing – review & editing, Data curation. **Andy Sheppard:** Writing – review & editing. **Rieks van Klinken:** Writing – review & editing. **Nianwan Yang:** Writing – review & editing, Funding acquisition. **Jue Wang:** Writing – review & editing. **Xiaoqing Xian:** Writing – review & editing, Investigation, Conceptualization. **Fanghao Wan:** Writing – review & editing. **Wanxue Liu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the National Key Research and Development Program of China [grant numbers 2023YFC2065200 and 2024YFC2607600], the Agricultural Science and Technology Innovation Program (ASTIP) (CAAS-ZDRW202505), and the Central Public-interest Scientific Institution Basal Research Fund (S2025XM07).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aiia.2026.07.004.

Data availability

The introduction risk network data of citrus pests (i.e., interception data from sources to entry ports), citrus yield data, distributions of the citrus pests, establishment and spread risk data, and codes of the introduction risk network analysis and ecological niche modeling process are openly available at Zenodo at https://zenodo.org/records/18290518. The environmental data sources are described in the Methods section.

References

Ali, H., Zhuo, Z., 2025. Integrated Pest Management: Emerging Technologies and Sustainable Approaches. first ed. CRC Press, Florida, USA.
Ali, H., Hou, Y., Tahir, M.B., 2023. Climate Change and Insect Biodiversity: Challenges and Implications. first ed. CRC Press, Florida, USA.
Araújo, M.B., New, M., 2007. Ensemble forecasting of species distributions. Trends Ecol. Evol. 22, 42–47.
BAAI, 2024. BGE-Reranker-v2. https://huggingface.co/BAAI/bge-reranker-v2-m3.
Bebber, D.P., Ramotowski, M.A.T., Gurr, S.J., 2013. Crop pests and pathogens move polewards in a warming world. Nat. Clim. Chang. 3, 985–988.
Bertelsmeier, C., Bonnamour, A., Garnas, J.R., Liu, T., Perreault, R., Ollier, S., 2025. Temporal dynamics and global flows of insect invasions in an era of globalization. Nat. Rev. Biodivers. 1, 90–103.
Blondel, V.D., Guillaume, J.L., Lambiotte, R., Lefebvre, E., 2008. Fast unfolding of communities in large networks. J. Stat. Mech. Theory Exp. 2008, P10008.
Bonacich, P., 2007. Some unique properties of eigenvector centrality. Soc. Networks 29, 555–564.
Bonnamour, A., Gippet, J.M.W., Bertelsmeier, C., 2021. Insect and plant invasions follow two waves of globalisation. Ecol. Lett. 24, 2418–2426.
Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32.
Capinha, C., Essl, F., Porto, M., Seebens, H., 2023. The worldwide networks of spread of recorded alien species. Proc. Natl. Acad. Sci. USA 120, e2201911120.
Chapman, D., Purse, B.V., Roy, H.E., Bullock, J.M., 2017. Global trade networks determine the distribution of invasive non-native species. Glob. Ecol. Biogeogr. 26, 907–917.
Chen, M., Tworek, J., Jun, H., Yuan, Q., Pinto, H.P.D.O., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., 2021. Evaluating Large Language Models Trained on Code. https://doi.org/10.48550/arXiv.2107.03374.

De Souza, A.A., Wang, N., Coletta-Filho, H.D., Del Ponte, E.M., 2020. Special issue on bacterial citrus diseases: part I. Trop. Plant Pathol. 45, 163–165.
Deutsch, C.A., Tewksbury, J.J., Tigchelaar, M., Battisti, D.S., Merrill, S.C., Huey, R.B., Naylor, R.L., 2018. Increase in crop losses to insect pests in a warming climate. Science 361, 916–919.
Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. Bert: pre-training of deep bidirectional transformers for language understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Association for Computational Linguistics, Pennsylvania, US, pp. 4171–4186.
Ding, D., Fu, X., Peng, X., Fan, X., Huang, H., Zhang, B., 2024. Leveraging chain-of-thought to enhance stance detection with prompt-tuning. Mathematics 12, 568.
Fenn-Moltu, G., Ollier, S., Bates, O.K., Liebhold, A.M., Nahrung, H.F., Pureswaran, D.S., Yamanaka, T., Bertelsmeier, C., 2023a. Global flows of insect transport and establishment: the role of biogeography, trade and regulations. Divers. Distrib. 29, 1478–1491.
Fenn-Moltu, G., Ollier, S., Caton, B., Liebhold, A.M., Nahrung, H., Pureswaran, D.S., Turner, R.M., Yamanaka, T., Bertelsmeier, C., 2023b. Alien insect dispersal mediated by the global movement of commodities. Ecol. Appl. 33, e2721.
Freeman, L.C., 1977. A set of measures of centrality based on betweenness. Sociometry 40, 35–41.
Froese, J.G., Murray, J.V., Beeton, N.J., van Klinken, R.D., 2024. The pest risk reduction scenario tool (PRReSTo) for quantifying trade-related plant pest risks and benefits of risk-reducing measures. Crop Prot. 176, 106484.
Githae, M.M., Coombes, C.A., Mutamiswa, R., Moore, S.D., Hill, M.P., 2025. Influence of overflooding ratios on fruit damage and population growth of *Thaumatotibia leucotreta* (Meyrick): implications for the sterile insect technique program. Pest Manag. Sci. 81, 7188–7196.
Gu, Y.J., Lu, X.F., Ding, G.L., Huang, F.Y., 2021. Pest interception from imported ASEAN fruit. Plant Quar. 35, 81–85.
Gullino, M.L., Albajes, R., Al-Jboory, I., Angelotti, F., Chakraborty, S., Garrett, K.A., Hurley, B.P., Juroszek, P., Makkouk, K., Pan, X., 2021. Scientific Review of the Impact of Climate Change on Plant Pests—A Global Challenge to Prevent and Mitigate Plant Pest Risks in Agriculture, Forestry and Ecosystems. FAO, Rome, Italy.
Gutierrez, A.P., Ponti, L., Neteler, M., Suckling, D.M., Cure, J.R., 2021. Invasive potential of tropical fruit flies in temperate regions under climate change. Commun. Biol. 4, 1141.
Hallouti, A., Ben El Caid, M., Boubaker, H., 2024. Mediterranean fruit fly *Ceratitis capitata* (Wiedemann) management strategies and recent advances: a review. Int. J. Pest Manag. 72, 157–169.
IPPC, 2025. IPPC Overview. https://www.ippc.int/en/about/overview/.
Jamieson, L.E., Woodberry, O., Mascaro, S., Meurisse, N., Jaksons, R., Brown, S.D.J., Ormsby, M., 2022. An integrated biosecurity risk assessment model (IBRAM) for evaluating the risk of import pathways for the establishment of invasive species. Risk Anal. 42, 1325–1345.
Karger, D.N., Conrad, O., Böhner, J., Kawohl, T., Kreft, H., Soria-Auza, R.W., Zimmermann, N.E., Linder, H.P., Kessler, M., 2017. Climatologies at high resolution for the earth's land surface areas. Sci. Data 4, 170122.
Karger, D.N., Conrad, O., Böhner, J., Kawohl, T., Kreft, H., Soria-Auza, R.W., Zimmermann, N.E., Linder, H.P., Kessler, M., 2021. Climatologies at High Resolution for the Earth's Land Surface Areas. EnviDat https://doi.org/10.16904/envidat.228.
Latora, V., Marchiori, M., 2001. Efficient behavior of small-world networks. Phys. Rev. Lett. 87, 198701.
Leroy, B., Delsol, R., Hugueny, B., Meynard, C.N., Barhoumi, C., Barbet-Massin, M., Bellard, C., 2018. Without quality presence–absence data, discrimination metrics such as TSS can be misleading measures of model performance. J. Biogeogr. 45, 1994–2002.
Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T., Riedel, S., 2021. Retrieval-augmented generation for knowledge-intensive NLP tasks. Adv. Neural Inf. Proces. Syst. 33, 9459–9474.
Li, H., Wu, H., Li, Q., Zhao, C., 2025. A review on enhancing agricultural intelligence with large language models. Artif. Intell. Agric. 15, 671–685.
Liu, C., White, M., Newell, G., 2011. Measuring and comparing the accuracy of species distribution models with presence-absence data. Ecography 34, 232–243.
Liu, C., Comte, L., Xian, W., Chen, Y., Olden, J.D., 2019a. Current and projected future risks of freshwater fish invasions in China. Ecography 42, 2074–2083.
Liu, C., Newell, G., White, M., 2019b. The effect of sample size on the accuracy of species distribution models: considering both presences and pseudo-absences or background sites. Ecography 42, 535–548.
Ma, C.S., Wang, B.X., Wang, X.J., Lin, Q.C., Zhang, W., Yang, X.F., van Baaren, J., Bebber, D.P., Eigenbrode, S.D., Zalucki, M.P., Zeng, J., Ma, G., 2025. Crop pest responses to global changes in climate and land management. Nat. Rev. Earth Environ. 6, 264–283.
Mani, M., Shivaraju, C., 2016. Mode of spread of mealybugs. In: Mani, M., Shivaraju, C. (Eds.), Mealybugs and their Management in Agricultural and Horticultural Crops. Springer, New Delhi, India, pp. 113–116.
Mathys, G., Smith, I.M., 1984. Regional and global plant quarantine strategies with special references to developments within EPPO. EPPO Bull. 14, 83–95.
Maxmen, A., 2013. Crop pests: under attack. Nature 501, S15–S17.
Mol, E.A.N., Kumar, M.B.S., 2024. End-to-end framework for agricultural entity extraction – a hybrid model with transformer. Comput. Electron. Agric. 225, 109309.
Montgomery, K., Walden-Schreiner, C., Saffer, A., Jones, C., Seliger, B.J., Worm, T., Tateosian, L., Shukunobe, M., Kumar, S., Meentemeyer, R.K., 2023. Forecasting global spread of invasive pests and pathogens through international trade. Ecosphere 14, e4740.
Moura, M.R., Oliveira, G.A., Paglia, A.P., Pires, M.M., Santos, B.A., 2023. Climate change should drive mammal defaunation in tropical dry forests. Glob. Chang. Biol. 29, 6931–6944.
Mu, H., Li, X., Wen, Y., Huang, J., Du, P., Su, W., Miao, S., Geng, M., 2022. A global record of annual terrestrial human footprint dataset from 2000 to 2018. Sci. Data 9, 176.
Nardi, D., Rosace, M.C., Cendoya, M., Marini, L., 2025. Trade network dynamics and alien plant pest introductions: a global analysis. Divers. Distrib. 31, e13963.
Niu, J.-Z., Hull-Sanders, H., Zhang, Y.X., Lin, J.Z., Dou, W., Wang, J.J., 2014. Biological control of arthropod pests in citrus orchards in China. Biol. Control 68, 15–22.
OpenAI, 2024. Text Embedding v3. https://platform.openai.com/docs/guides/embeddings.
Pacifici, K., Reich, B.J., Miller, D.A.W., Pease, B.S., 2019. Resolving misaligned spatial data with integrated species distribution models. Ecology 100, e02709.
Petitpierre, B., Broennimann, O., Kueffer, C., Daehler, C., Guisan, A., 2017. Selecting predictors to maximize the transferability of species distribution models: lessons from cross-continental plant invasions. Glob. Ecol. Biogeogr. 26, 275–287.
Phillips, S.J., Anderson, R.P., Dudík, M., Schapire, R.E., Blair, M.E., 2017. Opening the black box: an open-source release of Maxent. Ecography 40, 887–893.
Rafiq, K., Beery, S., Palmer, M.S., Harchaoui, Z., Abrahms, B., 2025. Generative AI as a tool to accelerate the field of ecology. Nat. Ecol. Evol. 9, 378–385.
Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W., Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40, 913–929.
Robertson, S., Zaragoza, H., 2009. The probabilistic relevance framework: BM25 and beyond. Found. Trends Inf. Retr. 3, 333–389.
Rosace, M.C., Conesa, D.V., López-Quílez, A., Marini, L., Martinez-Beneito, M.A., Nardi, D., Rossi, V., Vicent, A., Cendoya, M., 2025. Hotspot mapping of pest introductions in the EU: a regional analysis of environmental, anthropogenic and spatial effects. Biol. Invasions 27, 18.
Savary, S., Willocquet, L., Pethybridge, S.J., Esker, P., McRoberts, N., Nelson, A., 2019. The global burden of pathogens and pests on major food crops. Nat. Ecol. Evol. 3, 430–439.
Sun, P.S., Liu, M.D., Yan, J., Li, Z.H., Pan, X.B., 2017. Study on interrelation of plant quarantine pest lists. Plant Quar. 31, 15–21.
Tarusikirwa, V.L., Cuthbert, R.N., Mutamiswa, R., Nyamukondiwa, C., 2022. Context-dependent integrated stress resistance promotes a global invasive pest. Insect Sci. 29, 1790–1804.
Thuiller, W., 2024. Ecological niche modelling. Curr. Biol. 34, R225–R229.
Thuiller, W., Lafourcade, B., Engler, R., Araújo, M.B., 2009. BIOMOD – a platform for ensemble forecasting of species distributions. Ecography 32, 369–373.
Turner, R.M., Brockerhoff, E.G., Bertelsmeier, C., Blake, R.E., Caton, B., James, A., MacLeod, A., Nahrung, H.F., Pawson, S.M., Plank, M.J., Pureswaran, D.S., Seebens, H., Yamanaka, T., Liebhold, A.M., 2021. Worldwide border interceptions provide a window into human-mediated global insect movement. Ecol. Appl. 31, e02412.
Tzachor, A., Devare, M., Richards, C., Pypers, P., Ghosh, A., Koo, J., Johal, S., King, B., 2023. Large language models and agricultural extension services. Nat. Food 4, 941–948.
UN Comtrade, 2025. UN Comtrade Database. https://comtrade.un.org/.
Van Der Wal, J., Shoo, L.P., Graham, C., Williams, S.E., 2009. Selecting pseudo-absence data for presence-only distribution modeling: how far should you stray from what you know? Ecol. Model. 220, 589–594.
Van Klinken, R.D., Kingham, L., Hill, M.P., Collins, K., 2023. A menu of measures to manage trade-related plant pest risks, and a review of methods for demonstrating measure efficacy. Biol. Invasions 25, 1227–1248.
Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need. Adv. Neural Inf. Proces. Syst. 30. https://doi.org/10.48550/arXiv.1706.03762.
Wang, C., Wang, X., Jin, Z., Müller, C., Pugh, T.A.M., Chen, A., Wang, T., Huang, L., Zhang, Y., Li, L.X.Z., Piao, S., 2022. Occurrence of crop pests and diseases has largely increased in China since 1970. Nat. Food 3, 57–65.
Wang, L., Jin, T., Yang, J., Leonardis, A., Wang, F., Zheng, F., 2024. Agri-llava: Knowledge-Infused Large Multimodal Assistant on Agricultural Pests and Diseases. https://doi.org/10.48550/arXiv.2412.02158.
Warren, D.L., Matzke, N.J., Cardillo, M., Baumgartner, J.B., Beaumont, L.J., Turelli, M., Glor, R.E., Huron, N.A., Simões, M., Iglesias, T.L., Piquet, J.C., Dinnage, R., 2021. ENMTools 1.0: an R package for comparative ecological biogeography. Ecography 44, 504–511.
Wei, D., Dou, W., Wang, J.J., 2025. Theories and practices for integrated management of major citrus diseases and insect pests in China. Plant Prot. 51, 422–433.
World Trade Organization, 1994. The WTO Agreement on the Application of Sanitary and Phytosanitary Measures (SPS Agreement). World Trade Organization, Geneva, Switzerland.
Wu, G.A., Terol, J., Ibanez, V., López-García, A., Pérez-Román, E., Borredá, C., Domingo, C., Tadeo, F.R., Carbonell-Caballero, J., Alonso, R., Curk, F., Du, D., Ollitrault, P., Roose, M.L., Dopazo, J., Gmitter, F.G., Rokhsar, D.S., Talon, M., 2018. Genomics of the origin and evolution of citrus. Nature 554, 311–316.
Xiao, H., Chen, J., Chen, L., Chen, C., Wu, S., 2017. Exposure to mild temperatures decreases overwintering larval survival and post-diapause reproductive potential in the rice stem borer *Chilo suppressalis*. J. Pest. Sci. 90, 117–125.
Xu, Q., Gu, Y.J., 2023. Information analysis of quarantine scales from import products. J. Biosaf. 32, 126–132.
Yan, C., Liang, Z., Cheng, H., Li, S., Yang, G., Li, Z., Yin, L., Qu, J., Wang, J., Wu, G., Tian, Q., Yu, Q., Zhao, G., 2025. CDIP-ChatGLM3: a dual-model approach integrating computer vision and language modeling for crop disease identification and prescription. Comput. Electron. Agric. 236, 110442.
Ye, J., Lv, J.F., Liao, H., 2017. Situation analysis of the pest intercepted and countermeasures for the imported fruits in Guangdong during 2011-2016. Plant Quar. 31, 81–84.

ARTICLE IN PRESS

Yin, S., Fu, C., Zhao, S., Li, K., Sun, X., Xu, T., Chen, E., 2024. A survey on multimodal large language models. Natl. Sci. Rev. 11, nwae403.
Zhang, K., Ma, L., Cui, B., Li, X., Zhang, B., Xie, N., 2024. Visual large language model for wheat disease diagnosis in the wild. Comput. Electron. Agric. 227, 109587.
Zhao, H., Xian, X., Yang, N., Chen, T., Li, J., Sheppard, A., Wan, F., Qi, G., Liu, W., 2024a. A proposed coupling framework of biological invasions: quantifying the management prioritization in mealybugs invasion. Glob. Chang. Biol. 30, e17583.
Zhao, Y., Ji, C., Chen, Y., Zhu, X., 2024b. Who gains, who loses? – the impact of the belt and road initiative on bilateral agricultural trade. China Econ. Rev. 88, 102284.
Zhao, Z., Carey, J.R., Li, Z., 2024c. The global epidemic of bactrocera pests: mixed-species invasions and risk assessment. Annu. Rev. Entomol. 69, 219–237.